

u prošlom primeru jednostavno nismo imali dobar training dataset, bio je premali

- ✓ uvozimo fetch_20newsgroups dataset, vijesti koje želimo klasificirati

```
import pandas as pd

from sklearn.datasets import fetch_20newsgroups
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

```
categories = [
    "talk.politics.misc",
    "soc.religion.christian",
    "sci.space",
    "comp.graphics"
]

news = fetch_20newsgroups(
    subset="all",
    categories=categories,
    remove=("headers", "footers", "quotes")
)

print("Broj dokumenata:", len(news.data))
print("Nazivi kategorija:", news.target_names)
```

```
Broj dokumenata: 3732
Nazivi kategorija: ['comp.graphics', 'sci.space', 'soc.religion.christian', 'talk.politics.misc']
```

- ✓ Pregled dataseta

```
df = pd.DataFrame({
    "text": news.data,
```

```
0 \n\nPeter> : > Does anyone have any other suggestions where the 42 came from?\n\nPeter> Yep, here's a theory that I once heard bandied around.
1 In <May.13.02.31.16.1993.1569@geneva.rutgers.edu> djohnson@cs.ucsd.edu\n\n\nI'm afraid I see that question as very tendentious. Try rephrasing.
2 : Hi,\n\n: I am new to this newsgroup, and also fairly new to christianity.\n: ... I realize I am very ignorant about much of the Bible and
3 Does anyone out there have or know of, any kind of utility program for\n\nRibbons?\n\n\nRibbons are a popular representation for 2D shape.
4
```

	label
0	0
1	2
2	2
3	0
4	0

```
0 -> comp.graphics
1 -> sci.space
2 -> soc.religion.christian
3 -> talk.politics.misc
```

```
X_train, X_test, y_train, y_test = train_test_split(
    news.data,
    news.target,
    test_size=0.25,
    random_state=42,
    stratify=news.target
)
```

Bow + Logistic regression

```
# --- BAG OF WORDS ---
vectorizer_bow = CountVectorizer(stop_words="english", max_features=5000)

X_train_bow = vectorizer_bow.fit_transform(X_train)
X_test_bow = vectorizer_bow.transform(X_test)

# --- MODEL ---
model_bow = LogisticRegression(max_iter=1000)
model_bow.fit(X_train_bow, y_train)

# --- PREDIKCIJE ---
y_pred_bow = model_bow.predict(X_test_bow)

# --- REZULTATI ---
print("Accuracy with BoW:", accuracy_score(y_test, y_pred_bow))
print("\nClassification report with BoW:\n")
print(classification_report(y_test, y_pred_bow, target_names=news.target_names))
```

Accuracy with BoW: 0.8756698821007503

Classification report with BoW:

	precision	recall	f1-score	support
comp.graphics	0.89	0.89	0.89	243
sci.space	0.81	0.90	0.85	247
soc.religion.christian	0.95	0.86	0.90	249
talk.politics.misc	0.86	0.84	0.85	194
accuracy			0.88	933
macro avg	0.88	0.87	0.87	933
weighted avg	0.88	0.88	0.88	933

parametar max_features nam govori s koliko featurea radimo.

Manji broj - brži, jednostavniji model, ali može izostaviti ključne riječi

veći broj - sporiji, kompliciraniji, riskiramo overfitting

✓ TF-IDF + logistic regression

```
# --- TF-IDF ---
vectorizer_tfidf = TfidfVectorizer(stop_words="english", max_features=5000)

X_train_tfidf = vectorizer_tfidf.fit_transform(X_train)
X_test_tfidf = vectorizer_tfidf.transform(X_test)

# --- MODEL ---
model_tfidf = LogisticRegression(max_iter=1000)
model_tfidf.fit(X_train_tfidf, y_train)

# --- PREDIKCIJE ---
y_pred_tfidf = model_tfidf.predict(X_test_tfidf)

# --- REZULTATI ---
print("Accuracy with TF-IDF:", accuracy_score(y_test, y_pred_tfidf))
print("\nClassification report with TF-IDF:\n")
print(classification_report(y_test, y_pred_tfidf, target_names=news.target_names))
```

Accuracy with TF-IDF: 0.8981779206859593

Classification report with TF-IDF:

	precision	recall	f1-score	support
comp.graphics	0.91	0.91	0.91	243
sci.space	0.82	0.92	0.87	247
soc.religion.christian	0.96	0.90	0.93	249
talk.politics.misc	0.92	0.86	0.89	194
accuracy			0.90	933
macro avg	0.90	0.90	0.90	933

weighted avg	0.90	0.90	0.90	933
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```
print("BoW accuracy:  ", accuracy_score(y_test, y_pred_bow))
print("TF-IDF accuracy:", accuracy_score(y_test, y_pred_tfidf))
```

```
BoW accuracy:    0.8756698821007503
TF-IDF accuracy: 0.8981779206859593
```

```
print("BoW broj featurea:  ", X_train_bow.shape[1])
print("TF-IDF broj featurea:", X_train_tfidf.shape[1])
```

```
BoW broj featurea:    5000
TF-IDF broj featurea: 5000
```

✓ Test na novim tekstovima

```
new_articles = [
    "At the Franciscan youth gathering, the final football match was played between Gabriel's Angels and Ivan's Team. Officiated by Reverend",
    "The shuttle mission was delayed because of technical issues during launch preparation.",
    "The new graphics card improves rendering speed and image quality.",
    "The church community organized a large charity event for local families."
]

# BoW
new_X_bow = vectorizer_bow.transform(new_articles)
pred_bow = model_bow.predict(new_X_bow)

print("Predictions with BoW:\n")
for article, pred in zip(new_articles, pred_bow):
    print("TEXT:", article)
    print("PREDICTED:", news.target_names[pred])
    print("-" * 60)

# TF-IDF
new_X_tfidf = vectorizer_tfidf.transform(new_articles)
pred_tfidf = model_tfidf.predict(new_X_tfidf)

print("\nPredictions with TF-IDF:\n")
for article, pred in zip(new_articles, pred_tfidf):
```

```
print("TEXT:", article)
print("PREDICTED:", news.target_names[pred])
print("-" * 60)
```

Predictions with Bow:

TEXT: At the Franciscan youth gathering, the final football match was played between Gabriel's Angels and Ivan's Team. Officiated by Reverend
PREDICTED: soc.religion.christian

TEXT: The shuttle mission was delayed because of technical issues during launch preparation.
PREDICTED: sci.space

TEXT: The new graphics card improves rendering speed and image quality.
PREDICTED: comp.graphics

TEXT: The church community organized a large charity event for local families.
PREDICTED: soc.religion.christian

Predictions with TF-IDF:

TEXT: At the Franciscan youth gathering, the final football match was played between Gabriel's Angels and Ivan's Team. Officiated by Reverend
PREDICTED: soc.religion.christian

TEXT: The shuttle mission was delayed because of technical issues during launch preparation.
PREDICTED: sci.space

TEXT: The new graphics card improves rendering speed and image quality.
PREDICTED: comp.graphics

TEXT: The church community organized a large charity event for local families.
PREDICTED: soc.religion.christian

U oba slučaja koristimo isti ML model. Razlika nije u algoritmu, nego u reprezentaciji teksta. Bag-of-Words samo broji riječi, dok TF-IDF pokušava naglasiti riječi koje su informativnije za pojedini dokument.

Ako TF-IDF da bolji rezultat, to ne znači da je model pametniji — nego da smo mu dali bolju reprezentaciju podataka.

ZADATAK – RAD U GRUPAMA

Imate isti dataset i isti problem (klasifikacija teksta).

Vaš zadatak je napraviti **više verzija modela** i usporediti rezultate.

Što trebate napraviti

Napravite modele koristeći:

1. **Bag of Words (CountVectorizer)**
2. **TF-IDF (TfidfVectorizer)**

Za oba pristupa isprobajte:

- `ngram_range = (1,1)`
- `ngram_range = (1,2)`
- `max_features = 1000`
- `max_features = 5000`
- model:
 - `LogisticRegression`
 - `MultinomialNB`

Što trebate usporediti

Za svaki model pogledajte:

- accuracy
- razlike u rezultatima

Pitanja na koja morate odgovoriti

1. Koja kombinacija daje najbolji rezultat?
2. Je li TF-IDF uvijek bolji od BoW-a?
3. Kako `ngram_range` utječe na rezultat?
4. Kako `max_features` utječe na rezultat?
5. Je li veći model uvijek bolji?

Napomena

Ne tražimo savršen rezultat — tražimo objašnjenje. Najvažnije: **ZAŠTO se rezultat promijenio?**

Počnite pisati kôd ili generirati uz pomoć umjetne inteligencije.

